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Comparison of microburst-wind loads on low-rise structures of various geometric shapes

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ABSTRACT

Microburst can produce downdraft and strong divergent outflow wind, whose characteristics are distinct from the atmospheric boundary layer (ABL) wind. In the present study, microburst-wind loading effects on low-rise structures – a cubic-shaped building, a grain bin and two gable-roofed buildings – are evaluated and compared by performing laboratory tests on scaled models using a microburst simulator at Iowa State University. Velocity and turbulence intensity profiles at selected locations were studied. The distribution of mean and root-mean-square pressure coefficients for the models are shown for selected cases and compared with those obtained in the ABL wind. Results suggest that wind loads change significantly as the radial location, orientation and geometric shape of the structures vary. It was observed that at or near the center of the microburst, high external pressures occur for all structures, resulting in a large downward force on the roof. In the outburst region, the distribution of pressure coefficients on the structure envelope was found to be similar to those in the ABL wind, although actual wind load magnitudes may be much larger in the microburst wind. Different roof slopes and its cross-section resulted in different pressure distribution and overall wind loads on the models located in the outburst region. The low-angle gable roof and conical-shaped roof experience lower drag but larger uplift in the outburst region compared to buildings with flat roof and high-angle gable roof. The geometric parameters of the roof did not influence the wind loads at or near the center of the microburst where high positive static pressures govern the wind loads. Finally, it is found that the effect of geometric scale of a model on the mean wind loads in the outburst region is minor when it is within a blockage ratio of 3% as tested in the present study.

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1. Introduction

A microburst is a small-size and intense downdraft that impinges on the ground resulting in a divergent outburst wind with the radial extent being less than 4.0 km (Fujita, 1985). This damaging outburst wind can sometimes reach up to 168 mph (NOAA Website, 2010) with the maximum velocity very close to the ground surface. The flow field of a developed microburst is significantly different from that of the conventional atmospheric boundary layer (ABL) wind. Statistical summarization of the meteorological studies suggests that the flow field of a microburst at its maximum-wind producing status shares many similarities with the laboratory impinging jet flow (Hjelmfelt, 1988). Besides the wall-jet-like outburst flow, the microburst also produces high static pressure in the core and large turbulence in the divergent outflow. Because of these unique flow features, the microburst wind could be potentially dangerous to civil structures which are normally designed to resist the conventional ABL wind.

Low-rise structures, such as houses, grain bin or silos, industrial buildings and warehouses, etc., are common in rural and suburban areas of the United States. Low-rise buildings are particularly more vulnerable to extreme wind loads than engineered structures. Over past few decades, many studies have been conducted to investigate wind loading effects on various types of low-rise structures, through either wind tunnel tests or full-scale field studies. A limited number of citations are referred here because their data are used here for comparison (Castro and Robins, 1977; Cook and Redfearn, 1980; MacDonald et al., 1988). Due to the strong near-ground wind and unique flow characteristics, microburst winds are supposed to have distinct wind loading effects on low-rise structures. Current building codes and standards do not provide provision for estimation of wing loads of structures in a microburst for wind loads. Therefore, a better understanding of the microburst-wind loading effects is warranted, particularly in thunderstorm-prone areas. To date, several studies have been conducted to assess the microburst wind loads on cubic-shaped buildings. Nicholls et al. (1993) studied the flow structures around

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a cube-shaped house model in microburst-like winds. Chay and Letchford (2002) investigated the pressure distribution over a cube induced by a simulated microburst wind in a laboratory study. Sengupta and Sarkar (2008) conducted an experimental study to quantify the transient loads acting on a cube with an impinging-jet-based translating microburst simulator. Since the total number of related research is very limited, a lot of work is still needed to quantify the microburst-induced wind loads on different structures.

In the present study, the microburst wind loading effects on a set of low-rise building models have been investigated. The microburst wind was simulated by using a steady-impinging jet in WiST (Wind Simulation and Testing) Laboratory of the Department of Aerospace Engineering at Iowa State University. A steady impinging jet flow was used to simulate a steady microburst-like phenomenon which replicates the wind speed profile when the maximum wind is produced during the evolution of a microburst while ignoring the time-domain evolution of the microburst flow field. The models that were studied include a cubic-shaped building, a conical-roofed grain bin, and two gable-roofed buildings with different roof slopes but same plan dimensions. The purpose of this study was to establish the initial database of microburst wind loads for basic low-rise structures. By exploring the uniqueness and characteristics of the microburst wind loading, the authors hope that the results of the present study may help improve the wind loading design of the low-rise structures in thunderstorm-prone areas.

2. Experimental setup

Fig. 1 shows the steady impinging-jet flow simulator in the WiST Lab at Iowa State University and its schematic view, which was used in the present study. The jet flow is produced constantly by a fan on the top and impinges on a wooden ground plane to form a steady wall-jet flow field. The diameter of the nozzle (D) is about 0.6 m (2 feet). The geometric scale of the flow field was calculated as ~1:650 if compared to a small-sized microburst event with a diameter of 400 m (range 400–4000 m). The distance between the nozzle exit and the ground plane (H) was set to be 2 diameters of the nozzle ($H/D=2$). A honeycomb and several screens are placed at the exit of the nozzle to produce uniform velocity across the exit and reduce the turbulence of the issuing jet (approximately 2%). Velocity was measured using a Cobra probe (a multi-hole pressure probe, TFI Pvt. Ltd.®), which has the ability to measure all three velocity components simultaneously. The velocity data were sampled for 30 s at a sampling frequency of 1250 Hz at each measurement point, where the measurement uncertainty of the Cobra probe was within ± 0.5 m/s. The flow velocity at the nozzle exit of the ISU microburst simulator was set to 13 m/s (i.e., $V_{jet} \approx 13$ m/s). The corresponding Reynolds number of the flow was 5.2×10^5 based on the diameter of the jet nozzle. In Zhang et al. (2013), the velocity profile generated by this simulator was compared with existing data from the field and laboratory studies and reasonable agreement was found. Therefore, this steady impinging jet has proven to be a valid model for laboratory studies.

Fig. 2 presents the geometry of low-rise structural models used in the present study. All of these models were precisely fabricated using a 3D rapid prototyping machine located in the WiST Lab. To compare the effects of different roof shapes, the mean roof height of the grain bin model was kept the same as that of the gable-roofed models. Dimensions of all models and number of pressure taps are listed in Table 1. According to the geometric scale of 1:650 used here, the models studied represent large low-rise buildings like industrial buildings or warehouses instead of residential buildings. It is suggested that a larger microburst simulator be used in the future to test models with a larger geometric scale than used here to determine any scaling effects. The effects of scaling based on limited test cases are discussed later in this paper.

Pressure taps were uniformly distributed over these four models (see Table 1 for number of taps). These pressure taps were connected to DSA3217 pressure scanners (Digital Sensor Array, Scanivalve Corp.®) using tygon tubing (1.5 mm in diameter and 0.3 m long) for the surface pressure data acquisition. The pressure data were averaged over 10,000 data points collected with a frequency of 100 Hz. Since the tubing is trimmed equally short and no restrictors were included in the entire pressure acquisition system, the magnitude and phase distortion of the pressure fluctuation were insignificant (Irwin et al., 1979) and hence neglected in the present study. Both the velocity and pressure measurements were taken at five radial locations within the flow field, namely $r/D=0.0$, 0.5, 1.0, 1.5, and 2.0 as shown in Fig. 1. Resultant wind loads were also measured using a highly-sensitive force-moment sensor (JR3, model 30E12A-I40). The JR3 load cell is capable of measuring forces in three directions and the bending moment or torque about each axis. For each test run, 15,000 data points were taken with a sampling frequency of 1000 Hz. The measurement uncertainty of the sensor is $\pm 0.25\%$ of the full range (40 N).

3. Results and discussion

3.1. Velocity and turbulence intensity

The flow field of the steady impinging jet generally has greater complexity than the conventional ABL wind. The characteristics of

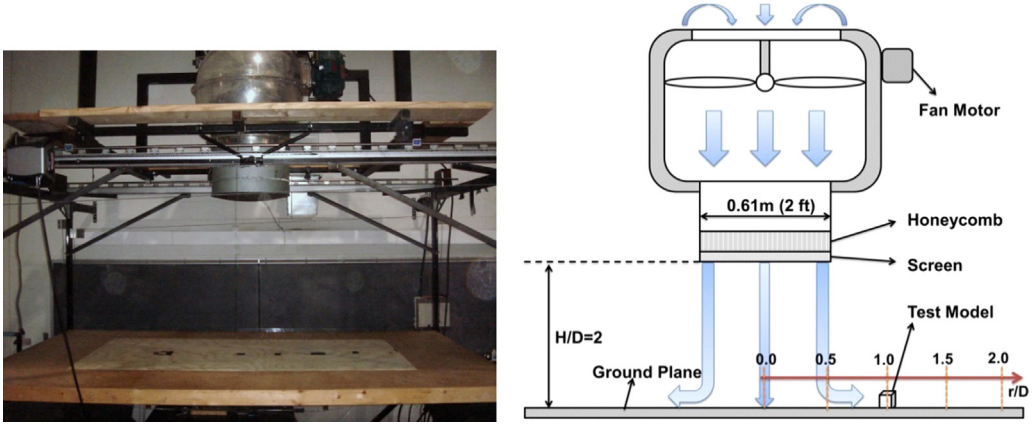


Fig. 1. Microburst simulator at Iowa State University: photo and schematic.

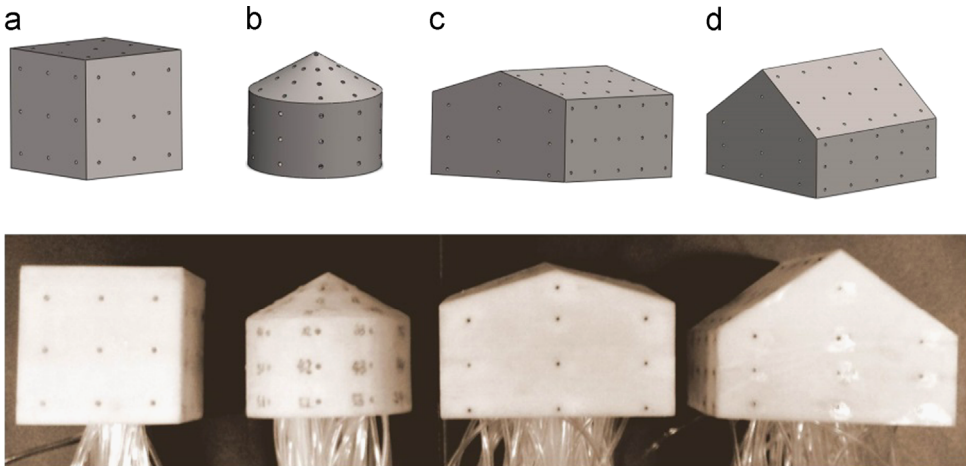


Fig. 2. Geometry of the building models. (a) Cube, (b) grain bin, (c) gable-roofed building (16 degree) and (d) gable-roofed building (35 degree).

Table 1
Dimensions of the low-rise structure models.

Dimensions	Cube	Grain bin	Gable-roofed building 16°	Gable-roofed building 35°
Mean roof height (mm)	45	36	36	36
Eave height (mm)	45	29	31	25
Total height (mm)	45	44	39	42
Roof angle (deg)	0	30	16	35
Base plan (mm)	45 × 45	Φ=50	65 × 65	65 × 65
Pressure tap numbers	45	61	80	80

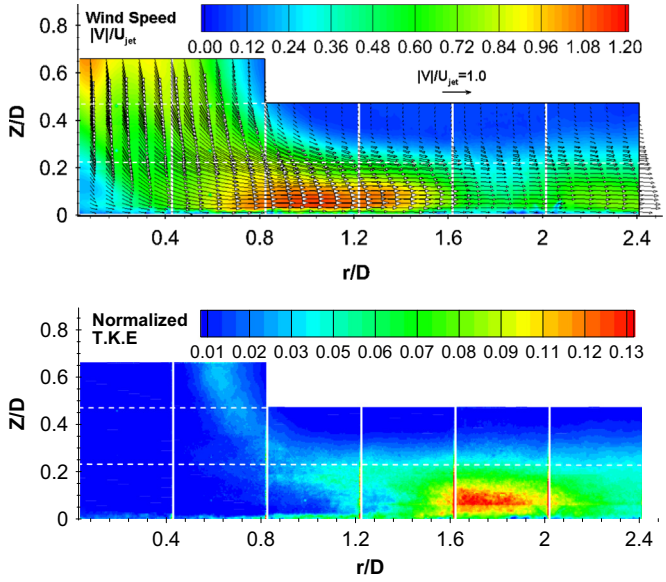


Fig. 3. Normalized velocity and turbulence kinetic energy (Zhang et al., 2013).

the current simulated microburst flow field have been discussed in detail in the author's previous study (Zhang et al., 2013). As shown in Fig. 3, the velocity and turbulence distributions show great variations spatially within the flow field. As a result, the surface static pressure also changes significantly along the radial direction as illustrated in Fig. 4. The entire picture of the microburst flow field can be depicted as follows. At the center of the microburst, a region of calm wind and high static pressure, namely the stagnation region, is formed due to the impact of the jet flow.

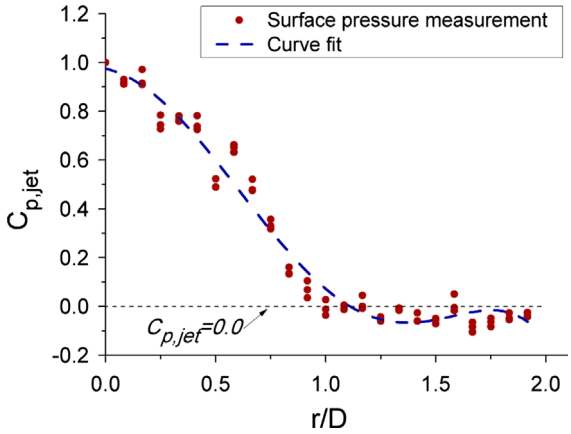


Fig. 4. Distribution of surface pressure coefficient.

From the center to approximately $r/D=1.0$, the effects of the static pressure field gradually diminish and the flow transitions from downdraft to radial outburst flow. At $r/D \geq 1$, which is named as outburst region in the present study, the radial velocity profile resembles a wall jet and the maximum velocity decreased significantly in radial direction.

The velocity and turbulence distributions at different radial locations are also measured by using point measurements in the present study. In Fig. 5, the radial velocity V_r was normalized by the same jet velocity ($V_{jet} \approx 13$ m/s) and the turbulence intensity was calculated by dividing the root-mean-square of local fluctuation by the local mean velocity. The normalized mean-roof-height of the building/grain bin models and the roof height of the cubic building are indicated in this figure as solid and dashed lines, respectively. It can be seen that the height of low-rise building models were much lower than the height of the shear layer ($z/D < 0.2$) where velocity and turbulence intensity changed dramatically. All the building models were immersed in the high outburst wind where velocity was uniformly high and turbulence was relatively low compared with those in the shear layer (at larger z/D). It should be noted that at $r/D=0.5$, the maximum radial velocity near the ground is much smaller ($V_r/V_{jet} \approx 0.62$) than those found in the outburst region. However, the radial velocity above $z/D=0.2$ is significant, which is a direct reflection of the transitioning flow from downdraft to radial outburst flow. The maximum radial velocity is found to be $V_r/V_{jet} \approx 1$ at $r/D=1.0$. The maximum radial velocity decreases slightly at $r/D=1.5$. At $r/D=2.0$, the local maximum radial velocity is much smaller

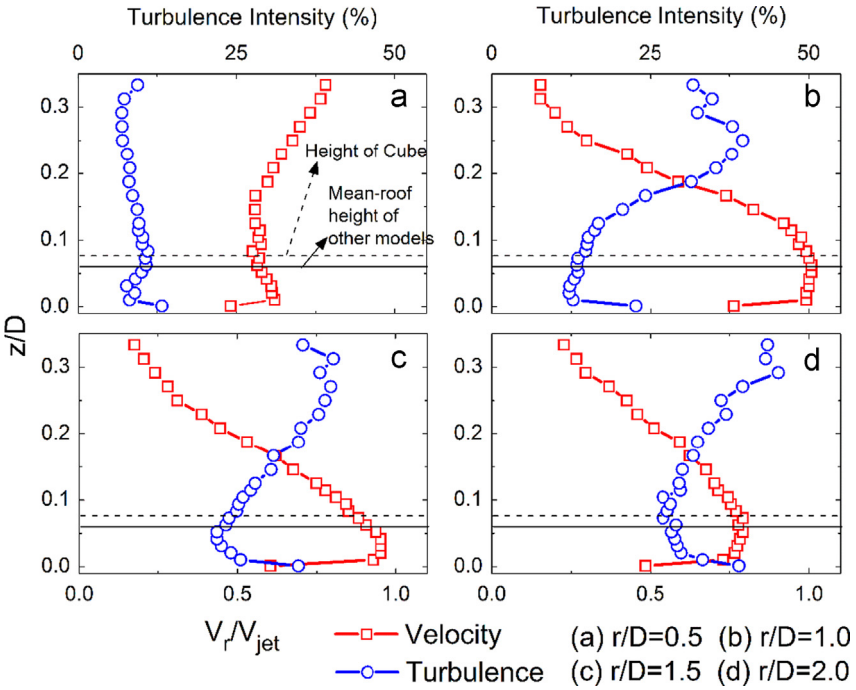


Fig. 5. Normalized radial velocity and longitudinal turbulence intensity profiles at different radial locations.

($V_r/V_{jet} \approx 0.8$) due to the turbulence mixing and dissipation. In general, the maximum radial velocity occurs at the mean-roof-height of the models or at a lower elevation. The local turbulence intensity at the mean-roof height of the buildings was found to increase monotonically from approximately 10% to 25% as the radial location changed from $r/D=0.5$ to 2.0. The maximum local turbulence intensity at a given normalized elevation z/D (normalized by the mean local radial velocity at this elevation) occurred at $r/D \approx 2.0$. However, the maximum turbulence intensity at the mean-roof-height, if normalized by the radial velocity at $r/D=2$ and at this height, occurred at $r/D=1.5$ since the radial velocity decreased significantly at $r/D=2.0$.

3.2. Mean and RMS pressure coefficients as a function of radial location

The mean pressure distribution on the building surfaces depend on the radial location of the building in the microburst flow field. The overall trend was found to be similar for all the low-rise building models and therefore only the contours for the grain bin models are shown here. In Fig. 6, the mean pressure distributions on the wall (expanded view) and on the roof are shown at different radial locations. The pressure coefficients were calculated as $C_{p,jet}=(p-p_{atm})/(0.5\rho V_{jet}^2)$, where V_{jet} is the jet velocity at the nozzle exit ($V_{jet} \approx 13$ m/s). It can be seen that at the center of the microburst, the pressure coefficient is uniformly ≈ 1.0 all around the grain bin model due to the local positive static pressure typical of stagnation region. At $r/D \approx 0.5$, though the surface pressure on the leeward side is reduced by half due to the flow-structure interaction, the entire surface experiences positive pressure. It indicates that the structure would experience considerable positive external pressure when the microburst occurs near the structure. This situation would be potentially dangerous for buildings, which are normally designed for roof uplift, and grain bins that are sealed for storage purpose. At $r/D \geq 1.0$, the pressure distributions were not significantly different from those found in the ABL wind (Cook and Redfearn, 1980; MacDonald et al., 1988) as discussed later. Related to change of the local wind speed and turbulence intensity, the magnitude of the positive pressure on the windward side of the wall and the negative pressure on roof and the side of the wall was found to decrease as r/D increased from 1.0 to 2.0.

Fig. 7 shows the root-mean-square of the fluctuating pressure coefficients at different radial locations. Generally, the pressure fluctuation is small when the grain bin model is at or near the center of the microburst, i.e. $r/D \approx 0.0$ and 0.5. At $r/D \geq 1.0$, high pressure fluctuations are found on the windward sides of the roof and side wall. The pressure fluctuation reaches the maximum when the model is at $r/D \approx 1.5$. At $r/D \approx 2.0$, the pressure fluctuation decreases slightly but is still larger than that found at $r/D \approx 1.0$. The result implies that the fluctuating surface pressure is directly linked to the fluctuation of wind speed at these radial locations. At $r/D \approx 2.0$, although the local turbulence intensity is higher than that at $r/D \approx 1.5$, the root-mean-square of the fluctuating velocity was actually smaller due to the decrease of mean velocity.

3.3. Comparison of wind loads on different structures

Low-rise building is a large category of civil structures and usually designed with various geometric shapes for different functions. Different geometric shapes trigger different types of flow-structure interaction and therefore require different standards for minimum wind load design. In microbursts, the geometric shape of the low-rise building will play an even more important role since the near-ground wind and turbulence are more significant than the conventional situation of the ABL wind. In this study, the four building models covered several key factors for low-rise building design, including different roofs (flat, conical, and gable) and cross-sections (square and circle).

Fig. 8 presents the comparison of the mean pressure distribution of different building models when they are at $r/D \approx 1.0$, where the maximum radial velocity is found to occur in the present study. First, it can be seen that the pressure distribution over building roofs varies considerably among different building models. For sharp-corner structures, such as the cube and the

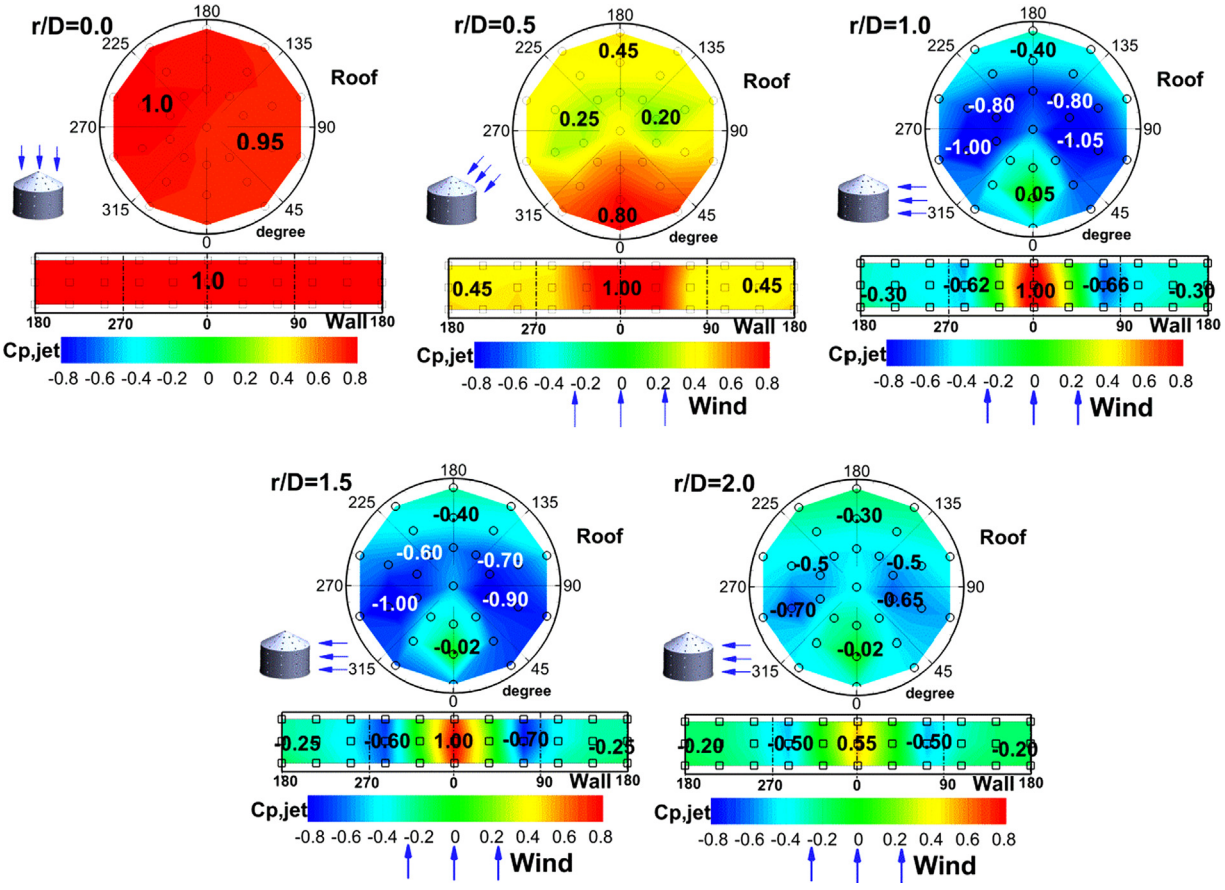


Fig. 6. Mean pressure coefficients for the grain bin model.

16° gable-roofed model, high suction (minimum negative pressure) is observed at the windward edges due to the local separation bubble caused by flow separation. For the conical roof of the grain bin model, a small positive pressure region is observed on the windward side along the centerline, which implies that no severe separation occurred at the leading edge with the larger roof angle (30°). However, a large negative pressure region is seen starting from approximately 45° with respect to the wind direction as shown in Fig. 8(b). The negative pressure region apparently covered larger portion of the roof than other models, indicating that the conical roof may experience more uplift force. Compared with 16° gable-roofed building, no negative pressure is seen at the windward edge of the 35-degree building. Pressure on the leeward side of the building is negative but distributed more uniformly implying the flow had completely separated over the roof ridge. Meanwhile, the circular cross-section of the grain bin model resulted in a different distribution on the wall. It can be seen that the area covered by high positive pressure on the wall is significantly reduced by the curvature of the grain bin model. The area covered by low negative pressure is also reduced on left and right sides of the circular wall compared with the other three models.

Fig. 9 shows the root-mean-square of pressure coefficient fluctuation of the four building models at $r/D \approx 1.5$, where the maximum turbulence intensity occurred (calculated on the basis of same wind speed and not local wind speed). Generally, the high RMS pressure region corresponded to those where large suction occurred as shown in Fig. 8. It is interesting to note that the pressure fluctuation on the roof of the cube is much smaller than that on the sidewalls in Fig. 9(a). This may be related to the decrease of velocity over the height of the cube as shown in Fig. 5(c). The 35-degree gable-roofed building model experienced the smallest pressure fluctuation on the roof, while the 16-degree gable-roofed building and grain bin models experienced large RMS pressure coefficients up to approximately 0.6 on the windward roof. The pressure fluctuations on the sidewalls of all models are observed to be significant. Although the magnitude of the negative pressure on the sidewall of the grain bin model is considerable, they are somewhat reduced because of the circular shape of the cross section. The maximum pressure fluctuation on the windward wall, which depends only on the oncoming flow turbulence, is found to be similar for all the four models.

3.4. Comparison with previous studies

The pressure coefficients along the centerline of the cube were extracted and compared for different radial location of the cube and with those of previous studies in Fig. 10. In Fig. 10(a), the centerline pressure distribution are compared for different radial locations where the pressure coefficients were calculated using the jet velocity as the reference velocity. Pressure distributions at $r/D \approx 0.0$ and 0.5 were completely different from those found in the outburst region, i.e. $r/D \approx 1.0$–2.0. At or near the center of the microburst, the local static pressure contributed to the positive external pressure over the building surface. In the outburst region ($r/D \geq 1.0$), the pressure coefficient distribution is found to be similar to those obtained in ABL wind. In Fig. 10(b), the pressure coefficient at $r/D \approx 1.0$ was compared with those found in the previous studies, including different types of wind, i.e. microburst and ABL winds. To validate this comparison, pressure coefficients were calculated using the eave height velocity (V_{eave}), i.e. $C_{p,jet} = (p - p_{atm})/(0.5\rho V_{eave}^2)$. It is found that the pressure on

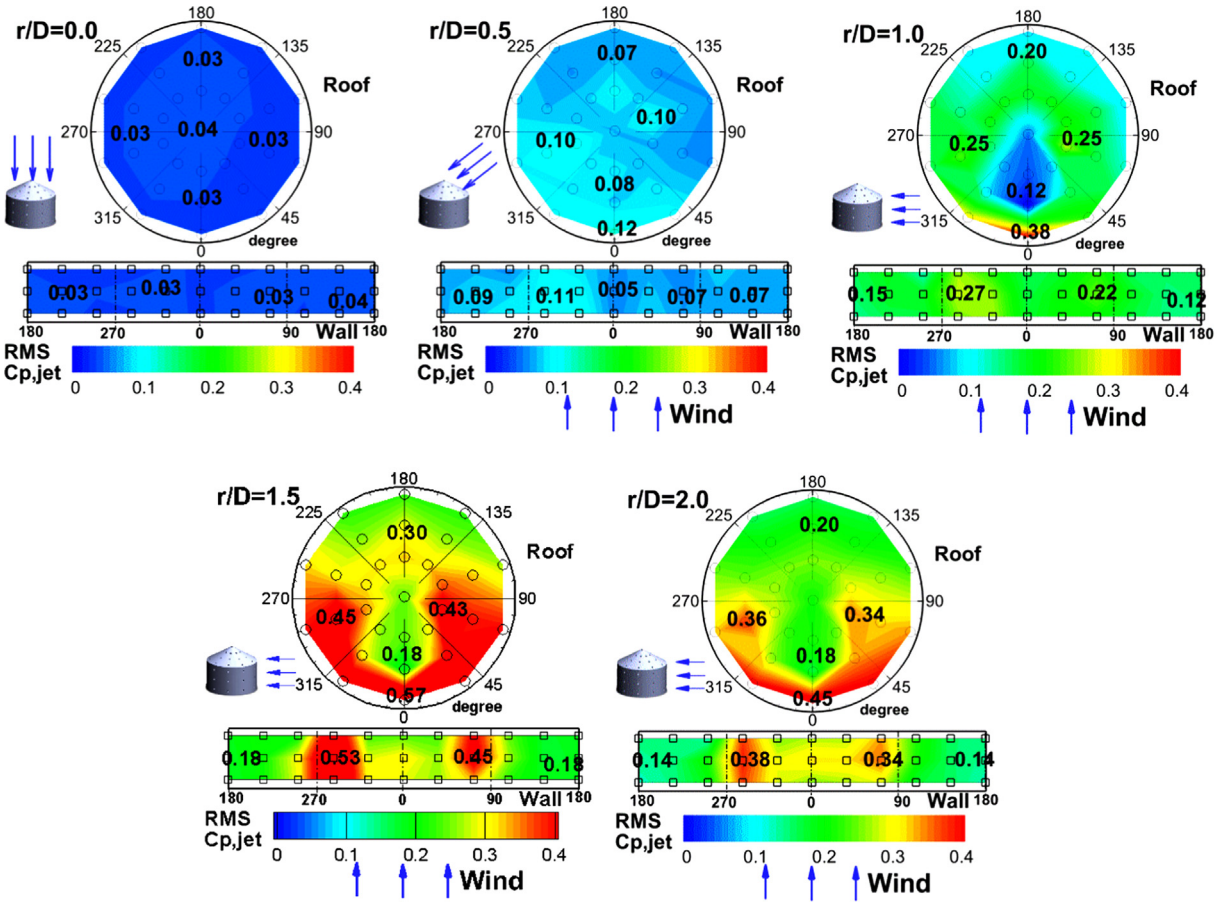


Fig. 7. Root-mean-square of fluctuating pressure coefficients for the grain bin model.

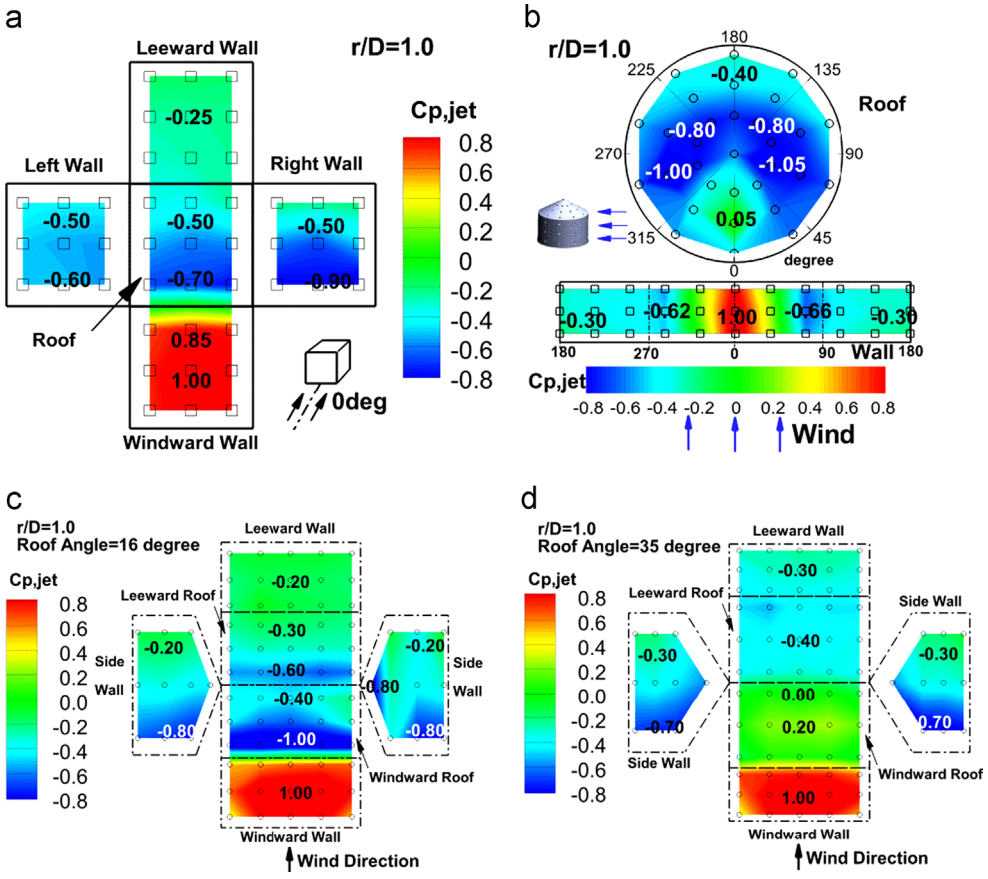


Fig. 8. Comparison of mean pressure coefficient at $r/D \approx 1.0$. (a) Cube, (b) grain bin, (c) gable-roofed 16 degree and (d) gable-roofed 35 degree.

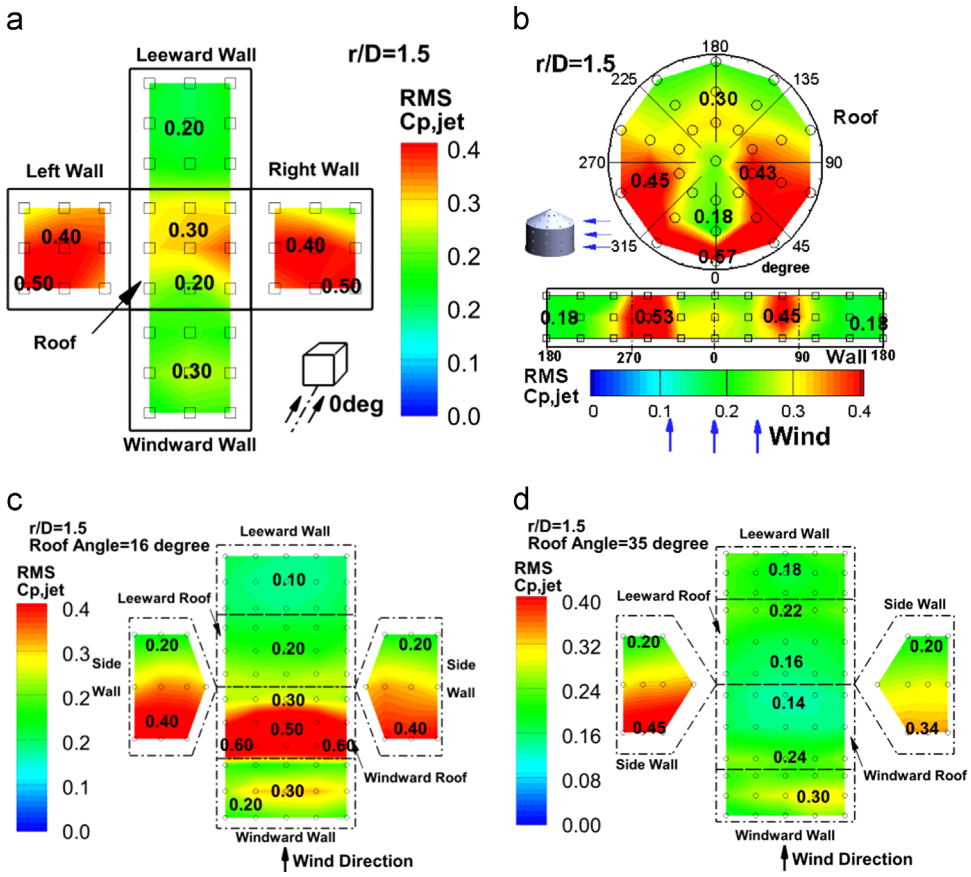


Fig. 9. Comparison of RMS pressure coefficient at $r/D \approx 1.5$. (a) Cube, (b) grain, (c) gable-roofed 16 degree and (d) gable-roofed 35 degree.

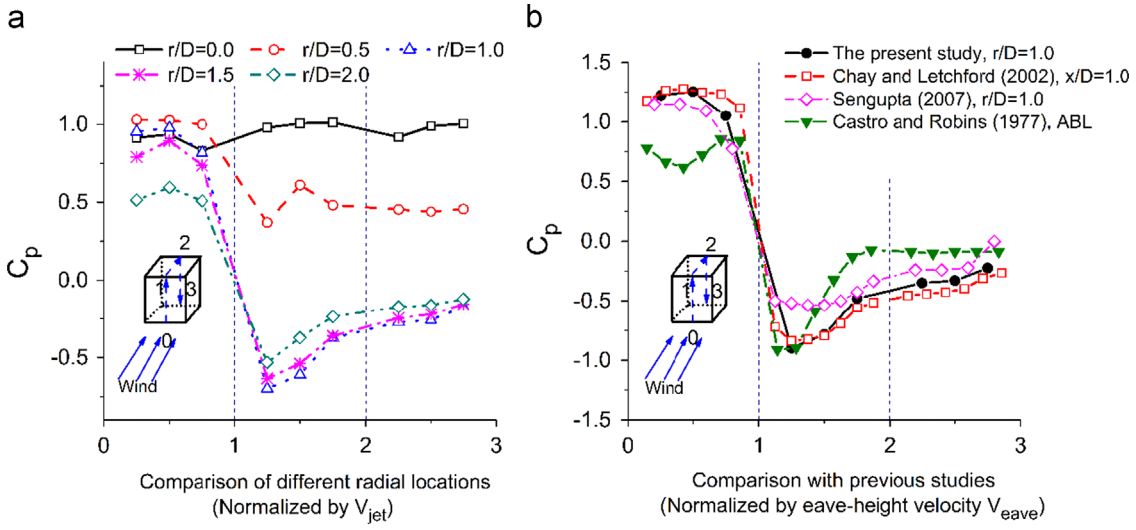


Fig. 10. Comparison of mean pressure coefficient along the centerline of the cube.

the windward wall induced by a microburst wind is higher than those expected in ABL wind because of the wall-jet-like wind profile. Among the results of microburst wind studies, a very good agreement is found between the present study and the previous studies of Chay and Letchford (2002) and Sengupta (2007). It should be noted that the jet height (H) to jet diameter (D) ratio in both of these studies was same as the present study, while the jet diameter (D) to model size (B) ratio used in these studies was slightly different, which were $D/B=17$ and $D/B=15.7$, respectively, compared to $D/B=13.3$ used here. This difference of blockage ratio (Plan area of model/Jet nozzle area) could be responsible for the slight differences found in this comparison apart from individual experimental errors. Further, it was suggested by Castro and Robins (1977) that the mean pressure distributions over the roof and leeward wall are closely related to the local turbulence intensity of the upstream flow, which was different in the aforementioned studies.

Available pressure data for the grain bin model was relatively scarce. The circumferential pressure distribution around the wall of grain bin silos at a height of 2/3 to 5/6 eaves height were documented in Cook and Redfearn (1980) and MacDonald et al. (1988), by taking full scale and wind tunnel measurements

respectively. The pressure data at $r/D \approx 1.5$, where maximum pressure fluctuation occurred, was extracted to compare with the data from the above studies in Fig. 11. The pressure was normalized again by the wind speed at eaves height of each model. As suggested by MacDonald et al. (1988), a factor of 1.35 was applied to the data of Cook and Redfearn (1980) since the pressure coefficient was calculated using wind speed at 10 m as the reference velocity. It can be seen that both the mean and RMS pressure coefficients followed the trends of the previous studies, particularly for the full-scale ABL wind study. The minimum mean pressure in the present study is found to be smaller in magnitude, while the maximum RMS of pressure fluctuation is found to be larger compared to the other studies. This discrepancy may result from the greater turbulence intensity in the present study, which was 22–25% at $r/D \approx 1.5$, compared with that found in MacDonald et al. (1988) (around 15–20%) and Cook and Redfearn (1980) (not specified).

The comparison of the pressure distribution on the gable-roofed building models was included in Zhang et al. (2013) and hence will not be discussed here. In conclusion, the mean and fluctuating pressure acting on the low-rise buildings in a micro-burst behaves quite differently from that in the normal ABL wind. At or near the center of the microburst, high static pressure plays an important role for the external pressure over the low-rise buildings. In the outburst region, the pressure distribution on the low-rise buildings is generally similar to that in the conventional ABL wind, though minor difference was still observed. These differences, including higher pressure on the windward wall and larger pressure fluctuations on the sidewalls, may result from the unique characteristics of the outburst flow, such as the wall-jet-shape wind profile and high turbulence intensity. It also should be noted that the actual pressure loadings on these buildings could be much higher than the standard design for ABL winds, as the pressure coefficients were calculated using different reference velocity.

3.5. Comparison of overall mean wind loads

The overall mean wind loads were calculated by integrating the pressure over the entire surface of building models. The mean radial and vertical force coefficients were then normalized by using the jet velocity and corresponding projection area as references. As seen in Fig. 12, both radial (drag, CF_r) and vertical (uplift, CF_z) direction force coefficients change significantly as a function of building geometry and radial location. For all models, the force coefficients change in a similar fashion as the radial location varies. At or near the center of the microburst, force along the radial direction acting on the low-rise structures was almost zero or relatively small. However, the downward pushing force acting on the roofs was much more significant. This unique situation induced by downdraft flow is usually not considered in the current design standards and therefore potentially dangerous for the safety of structures, particularly for a sealed structure. In the outburst region, i.e. $r/D \geq 1.0$, drag coefficient becomes more significant and vertical force coefficient becomes positive indicating uplift force on the roofs. The maximum drag and uplift for all models could be expected at $r/D \approx 1.0$, where the maximum velocity was seen within the microburst flow field.

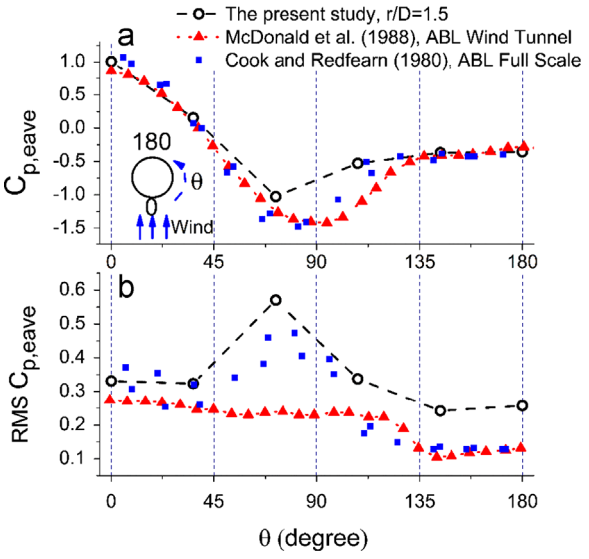


Fig. 11. Comparison of mean and RMS pressure coefficient along circumferences of the grain bin model (Reference velocity V_{eave}).

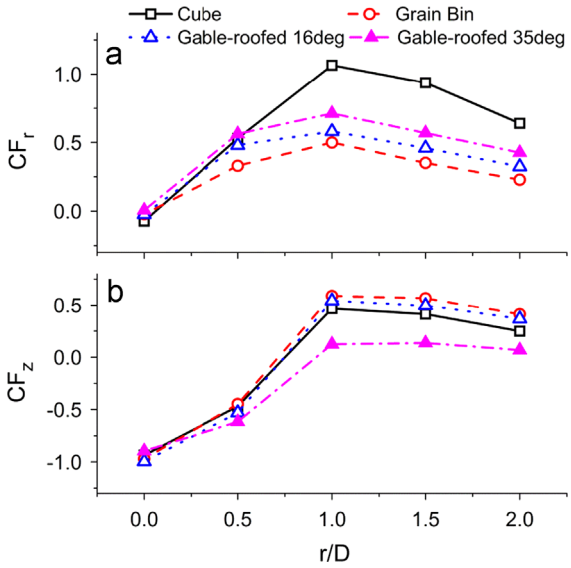


Fig. 12. Comparison of overall mean wind loads. (a) Radial and (b) vertical.

Building geometry also plays an important role in determining the overall mean wind loads. In Fig. 12, it can be seen that the difference among all tested models was not significant when they are at or near the center of the microburst. It implies again that the high static pressure, instead of the flow-structure interaction, has a greater contribution to the overall wind loads in these areas. In the outburst region, i.e. $r/D \geq 1.0$, the drag coefficient of the cube was the largest among all building models due to the large positive pressure on the windward wall (Fig. 8). The 35° gable-roofed building model experienced a larger drag than its 16° counterpart due to the blockage effect of the steeper roof. However, it is interesting to note that even though the roof angle of the grain bin model was not the smallest (30°), it actually experienced the smallest drag among all the structure models due to its circular cross-section, which corresponds well with the observation of the pressure distribution. Nevertheless, it can be seen, in Fig. 12(b), the uplift force coefficient acting on the conical roof of the grain bin was the largest, corresponding well with the large portion of suction area on the conical roof observed in Fig. 8(b). The 16° gable-roofed building and the cube also experienced large uplift in the outburst region due to the large suction area at the windward corner. The uplift force coefficient for the 35° gable-roofed building model was much smaller than those of other models corresponding well with the pressure distribution in Fig. 8.

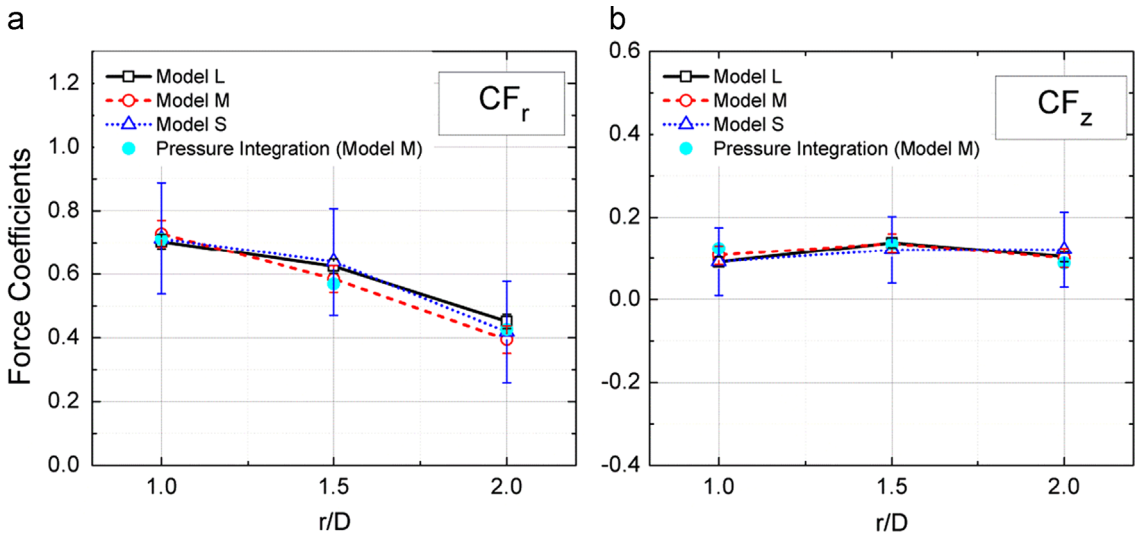


Fig. 13. Effects of model scale on mean wind loads in the outburst region (35-degree gable-roofed building), (a) radial-direction force coefficient and (b) uplift force coefficient.

3.6. *Model scale effects*

The effect of the model scale on microburst-wind loads has been investigated experimentally in this steady impinging jet flow. The 35-degree gable-roofed building model was used for this test. This model was duplicated with three sizes but the same geometric shape, namely Model L (91.8 mm × 91.8 mm (base plan) × 54 mm (mean-roof height)), Model M (65.3 mm × 65.3 mm × 36 mm), and Model S (32.7 mm × 32.7 mm × 18 mm).The blockage ratio of these three models, which is defined as the ratio between the base area of the models and the area of the issuing jet, were 2.9%, 1.5%, and 0.4%, respectively. The mean wind loads were measured using JR3 force balance in the outburst regions, i.e. $r/D=1.0$, 1.5, and 2.0, and compared with that obtained by pressure integration as discussed in previous sections. The drag and uplift force coefficients, as normalized by the model projection area normal to the direction of the force and jet velocity, are summarized in Fig. 13. Results suggest that the difference of mean drag and uplift coefficients among models with different sizes is minor in the outburst region. A good agreement is seen between the results obtained by pressure integration and by direct measurement using force balance. However, it should be noted that the measurement uncertainty was the largest for the smallest building model tested, as demonstrated by the error bars in Fig. 13. Therefore, it is suggested that models below a blockage ratio of 0.5% should be avoided in small-scale simulators with geometric scales of models below 1:200. Likewise, blockage ratio greater than 3% should be avoided in the absence of data from current and previous studies. Model scale effects with blockage ratio below 0.5% should be further studied with a larger simulator (5–7 times or larger) to ascertain the accuracy of very low blockage ratio that cannot be avoided for model studies of low-rise residential buildings (1–2 stories) with commonly used 1:100 to 1:150 scale.

4. Conclusions

A laboratory study was conducted to investigate the microburst-wind loading effects on low-rise buildings with various geometric shapes, including a cube, a grain bin model, and two gable-roofed building models. A steady impinging jet flow was employed generated by a microburst simulator located at Iowa State University. Velocity and turbulence intensity profiles were measured to reveal the complicated flow characteristics within the microburst-like wind. These flow field characteristics were then combined with the surface pressure measurements to show the microburst-wind loading effects on different building models, when they were mounted at different radial locations. Both the mean pressure and root-mean-square of the fluctuating pressure coefficients were studied and compared with those of the previous studies. Finally, the overall mean wind loads for all building models were summarized by integrating the surface pressure. The effects of different geometric shapes on the microburst-wind loading were summarized.

In conclusion, microburst-wind loading effects are more complicated than in normal ABL winds due to its complex flow regime. It is suggested that low-rise buildings would experience high external pressure and large downward force if a microburst occurs near the location of the building, such as $r/D \approx 0.0$ and 0.5 in the present study. In the outburst region, the distribution of pressure coefficients was basically similar to those obtained in ABL wind. However, some differences, such as higher mean pressure on the windward wall and higher pressure fluctuation on the sidewalls, are observed because of the wall-jet-like profile and high turbulence intensity of the outburst flow. Generally, the maximum mean pressure was found at $r/D \approx 1.0$ and the maximum pressure fluctuation was found at $r/D \approx 1.5$. Since the microburst winds are usually more violent than normal ABL winds, the magnitude of the actual pressure and force would be expected to be much more significant.

Geometric shape of the low-rise building also plays a key role, when the building models were in the outburst region. Pressure distributions suggested that flat, low-angle, and conical roof resulted in large suction areas and large pressure fluctuations on the roof due to the flow separation and reattachment. The circular cross-section of the grain bin model significantly reduced the area of high positive pressure on the windward side and negative pressure on the sidewall. Comparison of the overall mean wind loads revealed that the low-angle gable roof and conical roof resulted in reduced drag force on the structures when they were in the outburst region. In contrast, a low-angle gable roof and conical roof resulted in increased uplift force on the roof.

The effect of model scale was also studied here. The differences of mean wind loads in the outburst region among different models were minor for models tested within a blockage ratio of 3%. It is suggested that models below a blockage ratio of 0.5% and greater than 3% should be avoided in smaller-sized simulators (<2 m

nozzle diameter) to reduce the measurement uncertainty which limits the model studies to larger-sized structures with smaller simulators. To study low-rise structures with a more realistic geometric scale of 1:100 to 1:150, a larger microburst simulator (5–7 times the one used here) with models of less than 0.2% blockage ratio need to be used.

The goal of the present study is to establish a general understanding on microburst-wind loading effects on different low-rise structures. In future studies, it is suggested that entire life-cycle of the microburst flow field should be considered to resolve the time-domain variations of the microburst wind loading effects. A larger scale microburst simulator is also suggested to obtain better resolution of pressure measurement, particularly at regions with most research interests.

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